CS2601 Linear and Convex Optimization
Homework 7

Due: 2023.12.24

For this assignment, you should submit a **single** pdf file as well as your source code (.py or .ipynb files). The pdf file should include all necessary figures, the outputs of your Python code, and your answers to the questions. Do NOT submit your figures in separate files. Your answers in any of the .py or .ipynb files will NOT be graded.

1. **Lasso.** Implement the projection onto ℓ_1 ball. Use projected gradient descent to solve the Lasso problem on slide 12 of §11, you should complete the implementation of projected gradient descent method in `proj_gd.py`.

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \frac{1}{2}\|\boldsymbol{X}\boldsymbol{w}-\boldsymbol{y}\|_2^2 \\ \text{s.t.} \quad & \|\boldsymbol{w}\|_1 \le t \end{aligned} \qquad \boldsymbol{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \quad t = 1$$

Use the initial point $\boldsymbol{w}_0 = (0, 1.5)$ and 0.1 step size. Report the solution and the number of iterations. Plot the trajectory of $\boldsymbol{w}_k$ and the gap $f(\boldsymbol{w}_k) - f(\boldsymbol{w}^*)$.

2. Let $\boldsymbol{x} \in \mathbb{R}^3$. Consider

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) = e^{2x_1} + e^{x_2} + e^{x_3} \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \end{aligned} \tag{1}$$

(a). Solve problem (1) by the Lagrange multiplier method. Show the optimal solution $\boldsymbol{x}^*$, the Lagrange multiplier λ^* and the optimal value f^*.

(b). Find the closed-form expression for the Newton direction at a feasible $\boldsymbol{x}$ by solving the KKT system.

(c). Implement the constrained Newton's method on slide 15 of §12 in the `newton_eq` function of `newton.py`. The functions `numpy.block` and `numpy.linalg.solve` might be useful. Use your implementation to solve (1) with the initial point $\boldsymbol{x}_0 = (0, 1, 0)^T$. Show the output.

3. Consider the LP in standard form,

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \ge \boldsymbol{0} \end{aligned}$$

We are going to solve it using the barrier method.

(a). Write down the approximating equality constrained problem.

(b). Write down the gradient and Hessian matrix of the objective function you find in (a).

(c). Implement the barrier method for solving a generic LP in standard form with a given feasible initial point. Complete the functions `centering_step` and `barrier` in `LP.py`. For the centering step, i.e. line 3 on slide 14 of §13, you can use your implementation of constrained Newton's method in Problem 2(c) by defining the penalized objective function and its derivatives inside `centering_step`.

(d). Consider the following LP,

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & -x_1 - 3x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 8 \\ & -x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Convert it to standard form and then use your implementation in (c) to solve it. You can find a feasible initial point by setting $x_1 = 2, x_2 = 1$ and the slack variables appropriately. Show the output. Note `p3.py` also plots the projection of the iterates onto the x_1, x_2 coordinates.

4. Consider the LP in Problem 3(d).

(a). Find the dual LP in the standard form, i.e. with all four dual variables.

(b). Find the symmetric dual LP, i.e. without the dual variables for the primal nonnegativity constraints.

(c). Solve the dual LP in (b) graphically. Compare the dual optimal value and the primal optimal value computed in Problem 3(d).